

Classical Baselines and Feature Diagnostics

This report benchmarks fast classical tabular models before further ESN/QRC work.
All preprocessing, PCA, correlation pruning, thresholds, and model fitting are fit from train/validation only according to their role.
Primary model-selection metric shown here is validation PR-AUC because the stress class is imbalanced.
Test metrics are reported for diagnosis, but model choice should be justified from validation metrics.

Dataset summary

n_rows	n_columns	date_min	date_max	target	positive_rate	compact_features	expanded_features
7790	29	1993-04-27	2024-04-08	future_high_vol_label	0.20154043645699615	12	19

Split summary

split	n_rows	date_min	date_max	n_class_0	n_class_1	class_1_rate
train	5711	1993-04-27	2015-12-31	4516	1195	0.20924531605673263
val	1006	2016-01-04	2019-12-31	898	108	0.1073558648111332
test	1073	2020-01-02	2024-04-08	806	267	0.2488350419384902

Top classical baselines

feature_set	model_name	n_features	val_pr_auc	val_f1_class_1	test_pr_auc	test_f1_class_1
compact_corr_pruned_0.95	ridge_classifier	11	0.5777176960741521	0.5620437956204379	0.6610463369301105	0.5484581497797357
compact	ridge_classifier	12	0.5660137755754048	0.5592105263157895	0.6587587975354166	0.5212981744421906
expanded	logistic_l2	19	0.5515809815510889	0.5641025641025641	0.7102895413953512	0.5670225385527876
compact_pca_6	random_forest	6	0.5512045086984461	0.549618320610687	0.6669757115423633	0.6053639846743295
expanded	linear_svc	19	0.5507248004486996	0.5605536332179931	0.7078033607256693	0.5537757437070938
expanded	logistic_l1	19	0.5488891940009739	0.5627376425855514	0.7129687173670989	0.5799522673031027
compact_pca_6	extra_trees	6	0.5480932352200661	0.5583333333333333	0.6923362651990225	0.5992010652463382
expanded	ridge_classifier	19	0.5464924123329175	0.5567010309278351	0.6430553888091572	0.6267716535433071
compact_corr_pruned_0.95	logistic_l1	11	0.5428216717375737	0.5645933014354066	0.705523331855711	0.6366366366366366
compact	logistic_l1	12	0.5427815328265285	0.5645933014354066	0.7054353761565356	0.6366366366366366
compact	logistic_l2	12	0.541754617373711	0.5741626794258373	0.7042540454599536	0.6336336336336337
compact	linear_svc	12	0.5416080965650655	0.5727699530516432	0.701681320449918	0.6350148367952523
compact_corr_pruned_0.95	logistic_l2	11	0.5408186883452422	0.5692883895131086	0.7042807980061171	0.5795724465558195
compact_corr_pruned_0.95	linear_svc	11	0.540221681600051	0.5674418604651162	0.7024315898896053	0.6328358208955224
compact_pca_6	linear_svc	6	0.5254291167694737	0.5351170568561873	0.7068104961132865	0.5607264472190693
compact_pca_6	logistic_l2	6	0.5201022145587163	0.5338983050847458	0.7079871360478415	0.627939142461964
compact_pca_6	logistic_l1	6	0.5197707122590501	0.5342960288808665	0.7078963219359736	0.5710928319623971
compact_pca_6	ridge_classifier	6	0.5144450957364677	0.5223367697594502	0.7051146810344737	0.5594713656387665

Highest feature correlations

feature_1	feature_2	abs_corr
vix_close	vix_ma_5d	0.9797709789081838
rv_10d	rv_20d	0.9281048176665099
vix_change	vix_pct_change	0.9056799566094704
rv_5d	rv_10d	0.9016476633012773
rv_20d	vix_ma_5d	0.8999163972204771
rv_20d	vix_close	0.8763661409753596
rv_10d	vix_ma_5d	0.8712940823681149
rv_10d	vix_close	0.8557361189135868
rv_5d	rv_20d	0.8185515120769222
rv_5d	vix_ma_5d	0.7941262209990385
spy_log_return	vix_change	0.7940259282991708
rv_5d	vix_close	0.7879896281080239
spy_range	vix_close	0.7586142892877107
spy_range	rv_5d	0.7241552790062799
spy_range	rv_10d	0.7232001214932108

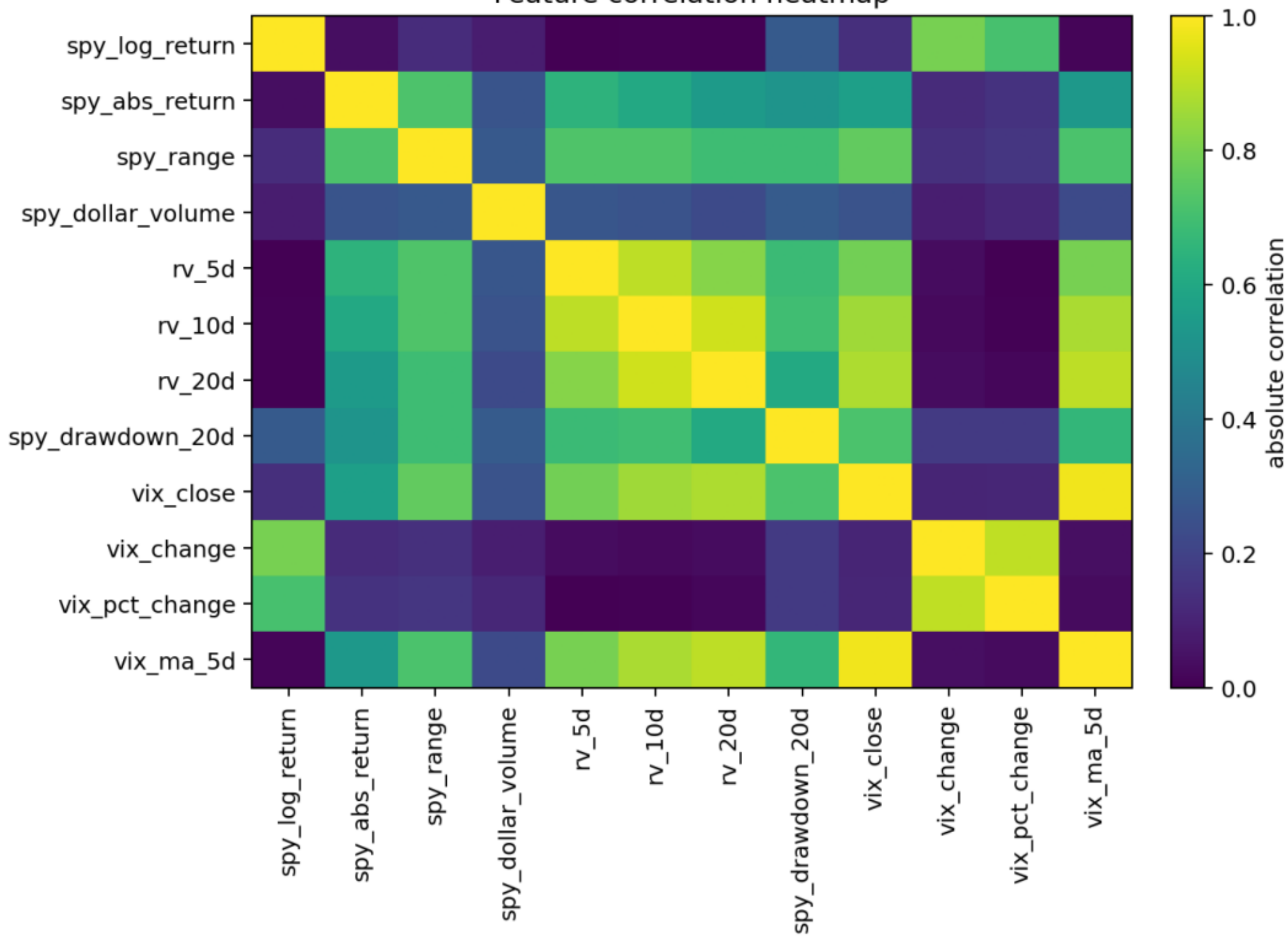
PCA explained variance

component	explained_variance_ratio	cumulative_explained_variance
1	0.5195221710707585	0.5195221710707585
2	0.22431919970363084	0.7438413707743894
3	0.07534561804305977	0.8191869888174491
4	0.05816816244913522	0.8773551512665844
5	0.03956509052403656	0.9169202417906209
6	0.025314247012172116	0.942234488802793

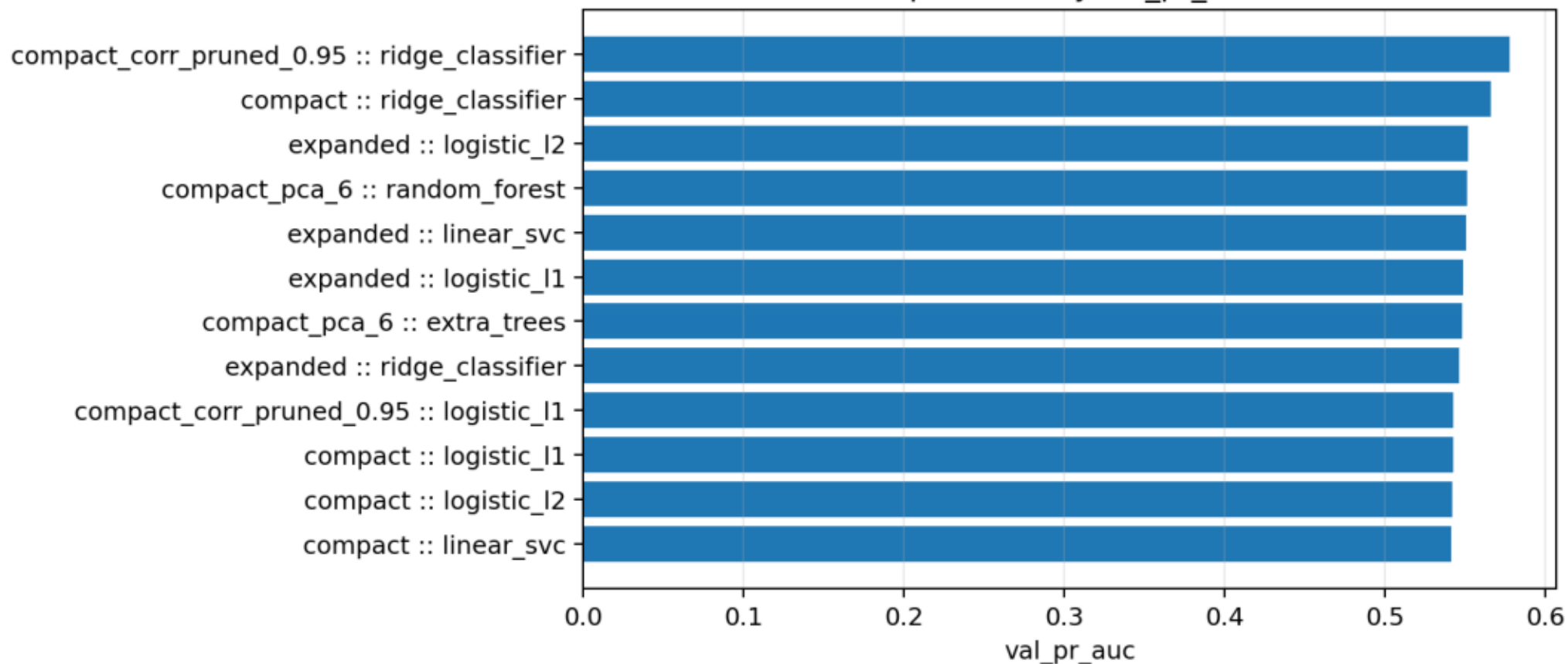
Best-model permutation importance

feature	importance_mean	importance_std
vix_close	0.4543415204102331	0.01165294574462852
vix_pct_change	0.128327556389715	0.013749648617216685
vix_change	0.09642466587603435	0.01503892074502625
rv_10d	0.08764815622836289	0.018836014042847977
rv_20d	0.06711083822877284	0.017638499859983593
spy_range	0.010411008396741455	0.006050209638679077
rv_5d	0.0074468960048021445	0.0034692551579176903
spy_abs_return	0.0023498637645842425	0.003276222058533641
spy_drawdown_20d	0.0010794876895150907	0.0009314319126935351
spy_dollar_volume	-0.0014398538735155176	0.001969162760621574
spy_log_return	-0.0026229720423299905	0.004785708602689463

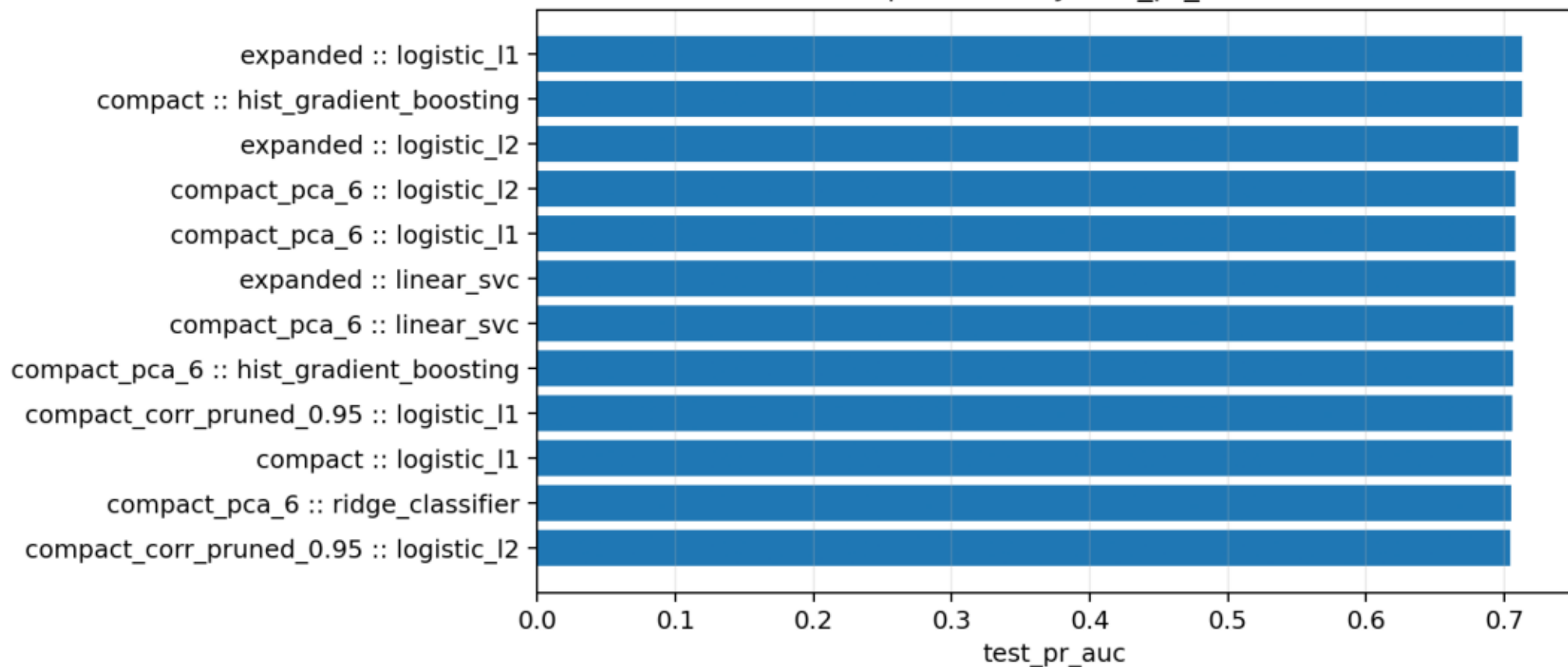
Feature correlation heatmap



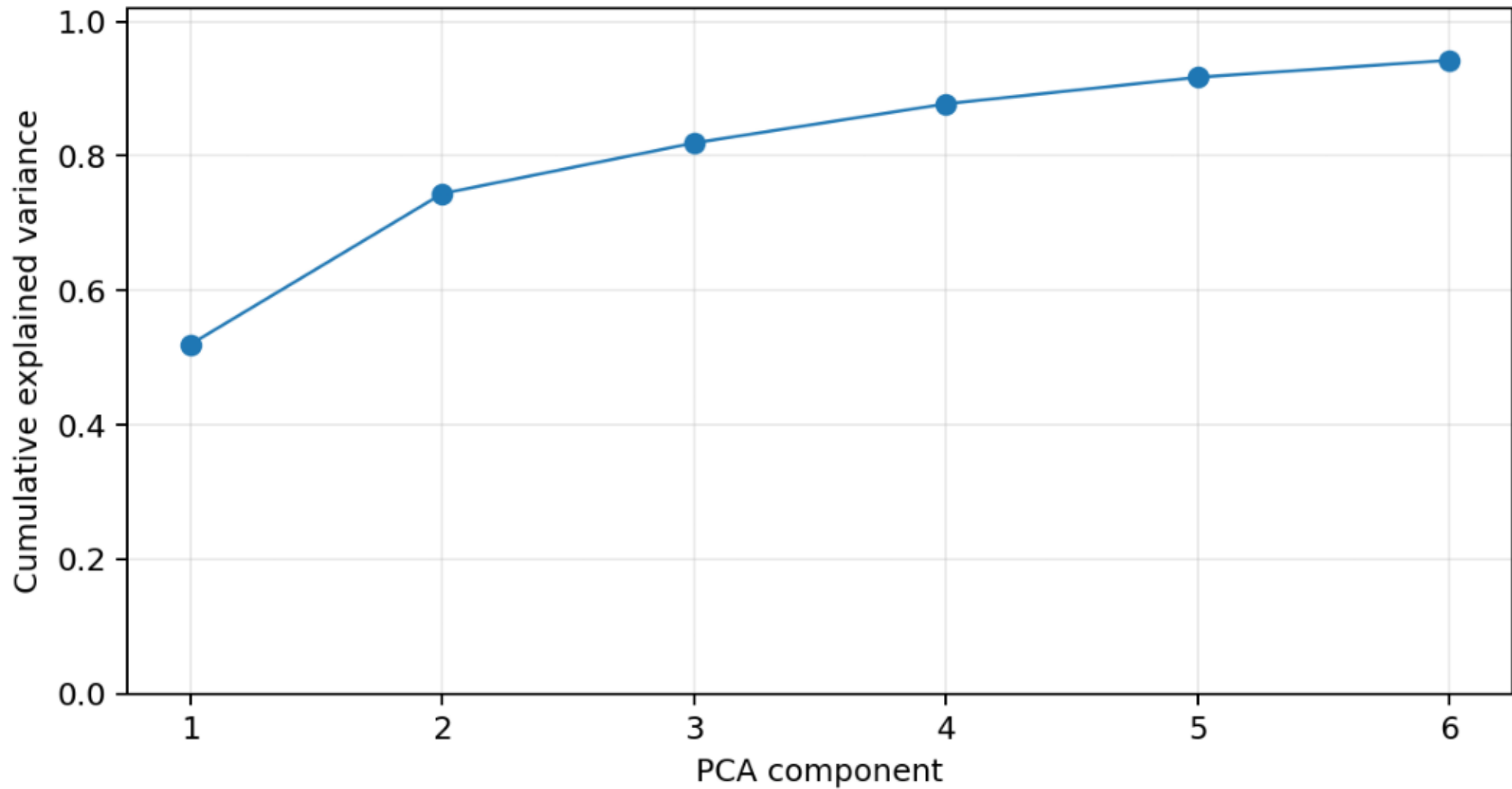
Top models by val_pr_auc



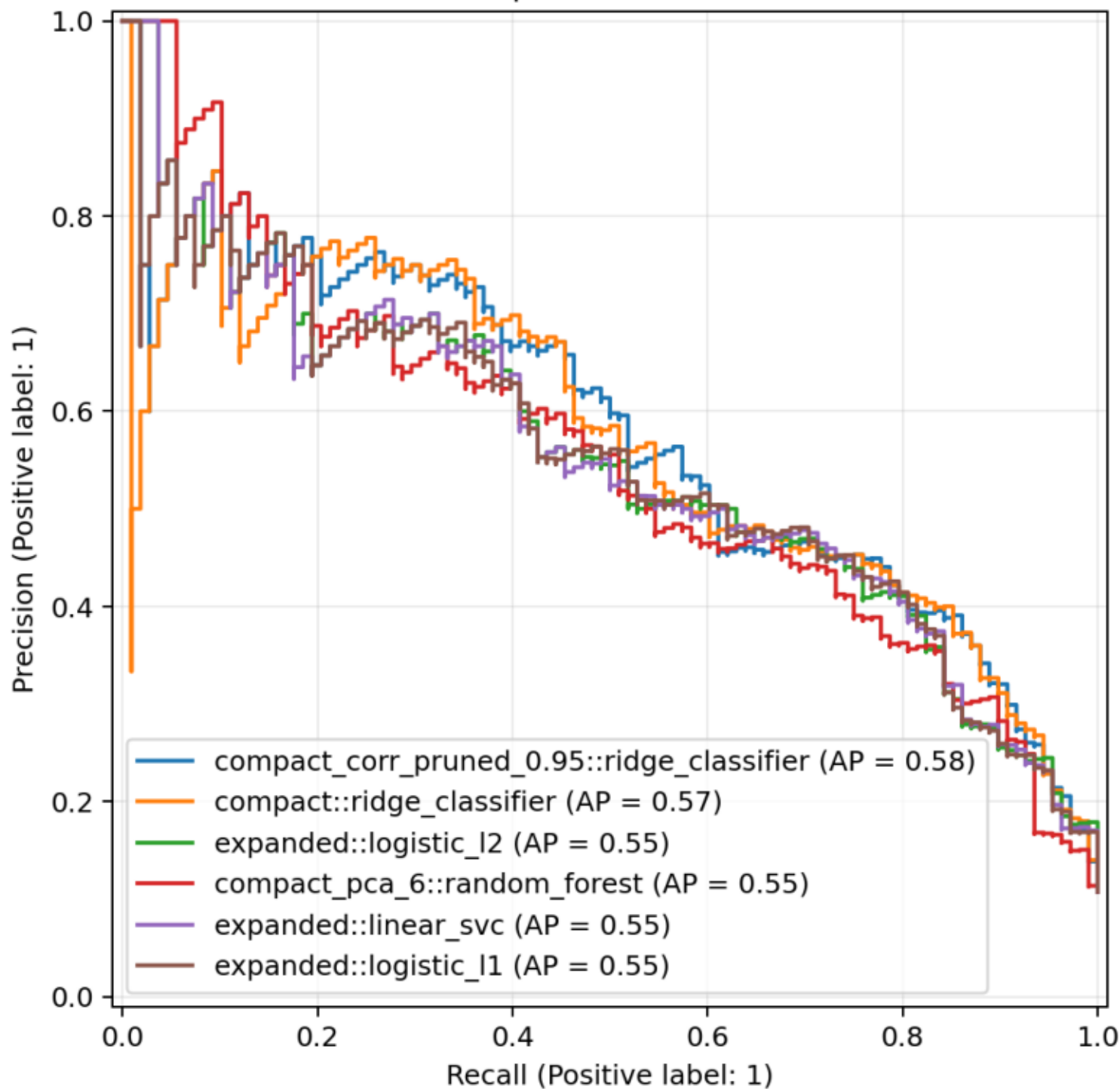
Top models by test_pr_auc



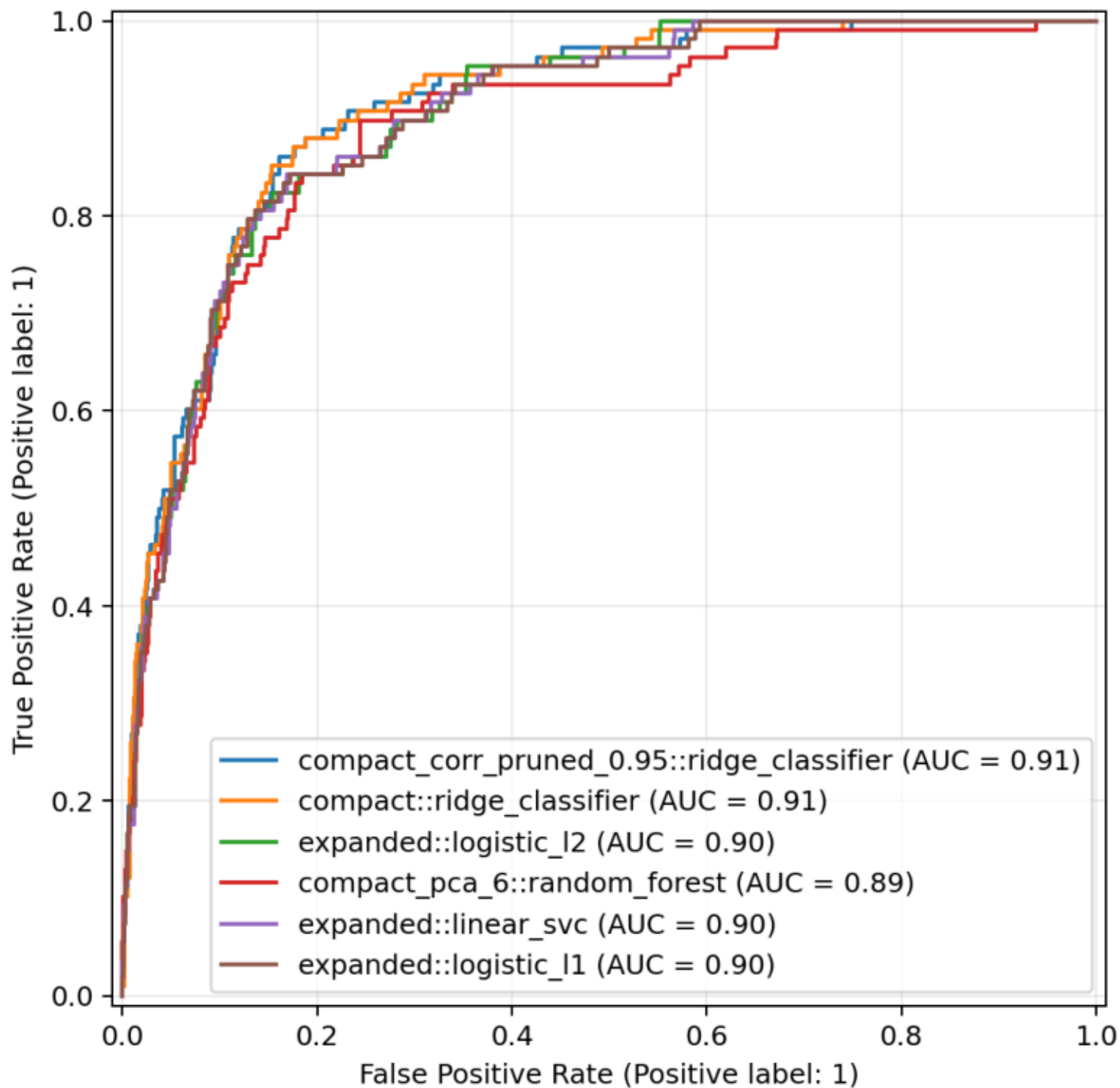
PCA cumulative explained variance



Validation precision-recall curves



Validation ROC curves



Permutation importance: compact_corr_pruned_0.95::ridge_classifier

